HERSAFAR
A VOICE-FIRST COMMUTE APP

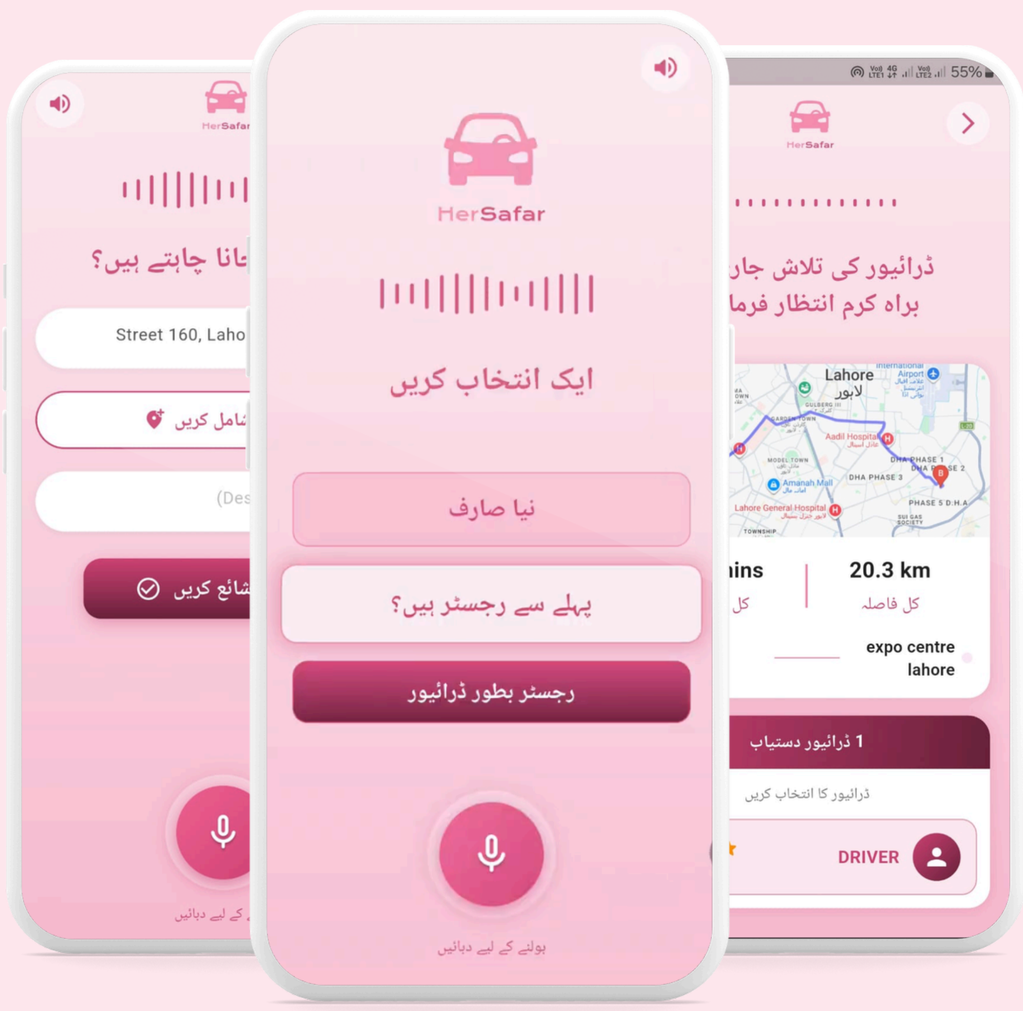
Muhammad Zain Naseer, Muhammad, Kousar Pervaiz, Maryam Rizwan, Emaan Waleed
Advisor: Waqar Ahmed

Problem

Female domestic workers in Pakistan face inaccessible, unsafe, and costly commuting. Digital mobility apps are unusable due to language and literacy barriers, while public transport is overcrowded and unpredictable and private options are unaffordable, leaving them without safe, affordable, and accessible travel support.

Objectives

- Enable rides for female domestic workers
- Accessible voice–first, Urdu interface
- Economic affordability via optimized routing and carpooling
- Built–in SOS and real–time safety features



System Architecture

Client Layer	API Gateway	Services Layer	Data Layer
Flutter App (IOS/Android)	**Node.js + Express Rest API**	**Gemini API NLU + TTS**	**MongoDB Atlas Cloud Database**
Voice Input (STT) + Audio Output (TTS) Google Maps SDK Live Map UI	Socket.IO Real–Time Events Auth & Role Middleware	Ride Matching Engine SOS Alert Service	User Profiles Ride History Safety Logs + Reports

Core System Features

- **Voice–First Navigation**
- **Emergency SOS**
- **Live GPS Tracking**
- **Admin Safety Panel**
- **Role Based Flows**
- **Real–Time Updates**
- **Issue Reporting**
- **Secure Auth**

Future Work

- **AI–Powered Routing**
  ML models for demand–aware route optimization and smarter driver dispatch
- **Automated Data Extraction**
  Using Computer Vision to auto–fill profiles from scanned documents
- **Expanding Local Dialects**
  Scaling voice support for more regional languages